nature catalysis
SUPPLEMENTARY INFORMATION
https://doi.org/10.1038/s41929-020-0476-3
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Molecular-level insights on the reactive facet of carbon nitride single crystals photocatalysing overall water splitting

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Molecular-level insights on the reactive facet of carbon nitride single crystals photocatalyzing overall water splitting

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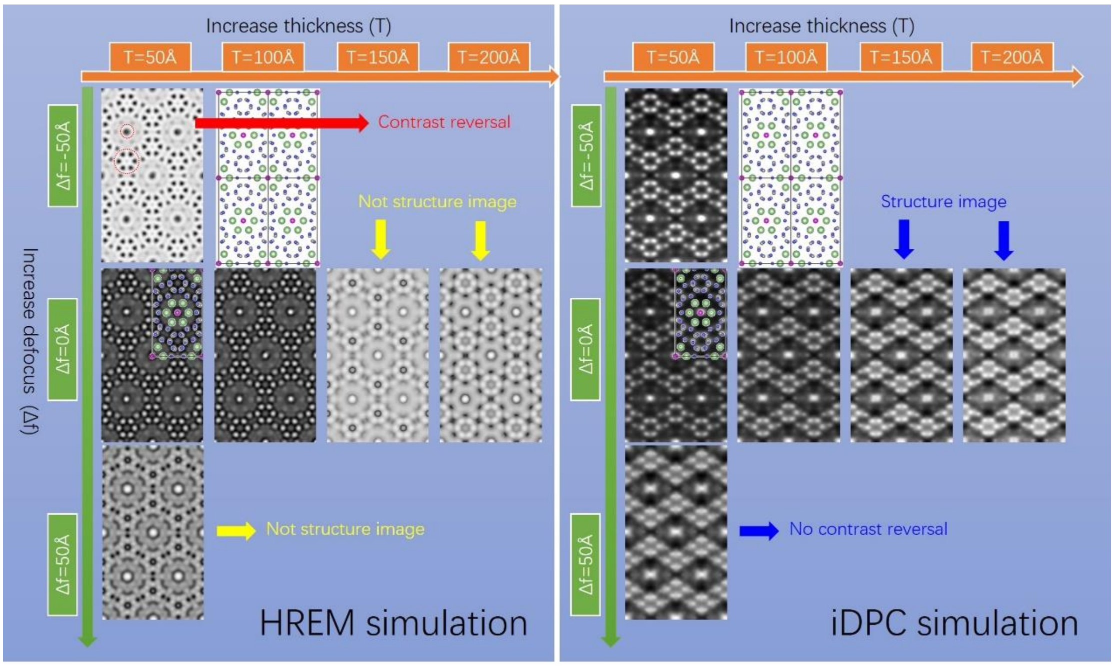
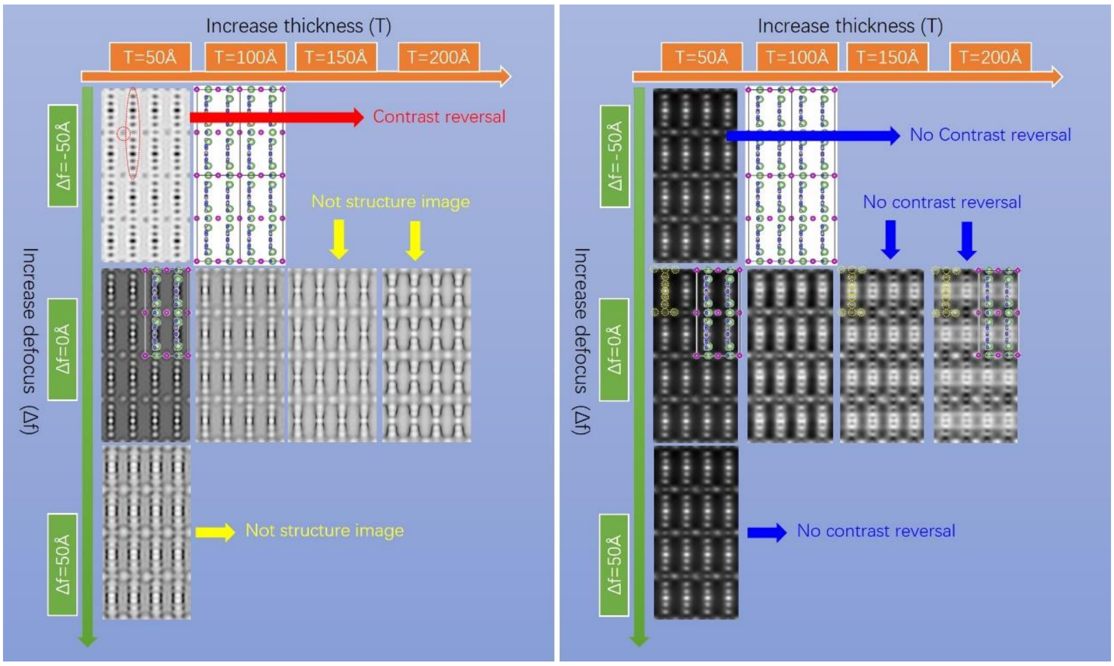
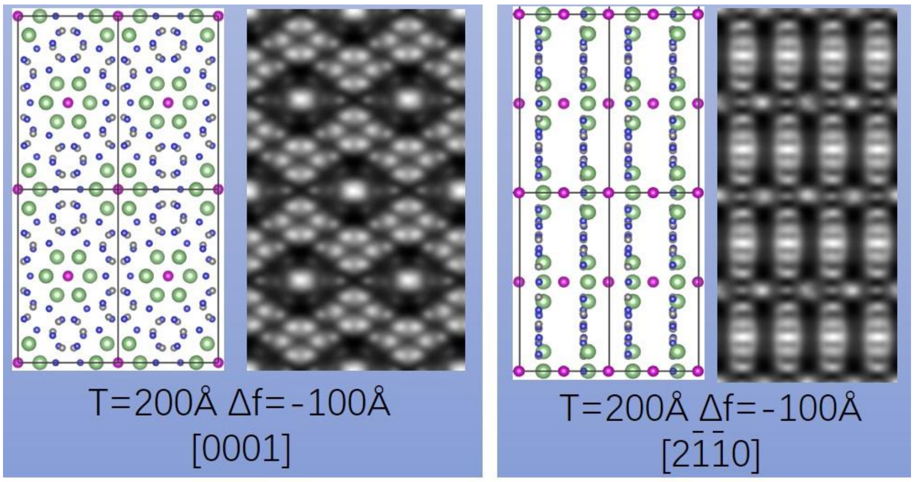


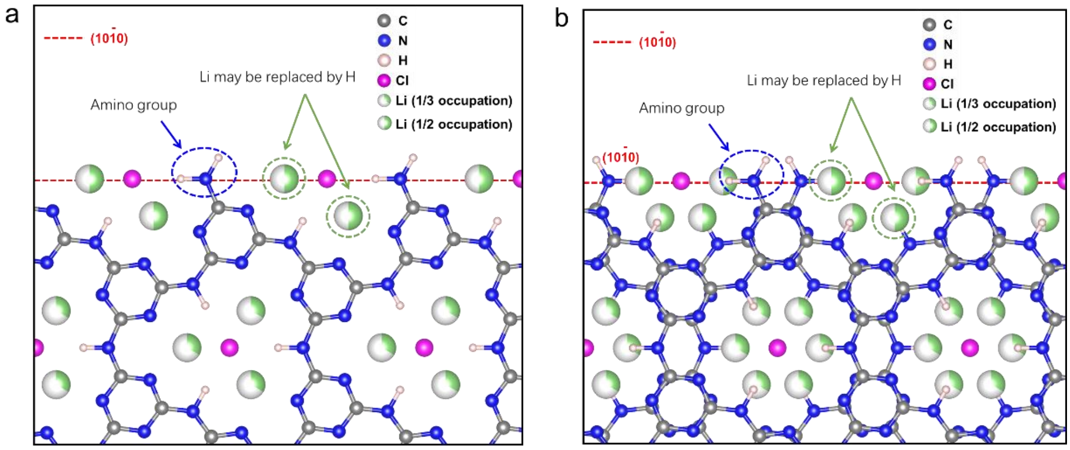
Supplementary Figure 1. HREM (left pane) and iDPC (right panel) image simulation of PTI crystals along the [0001] direction.



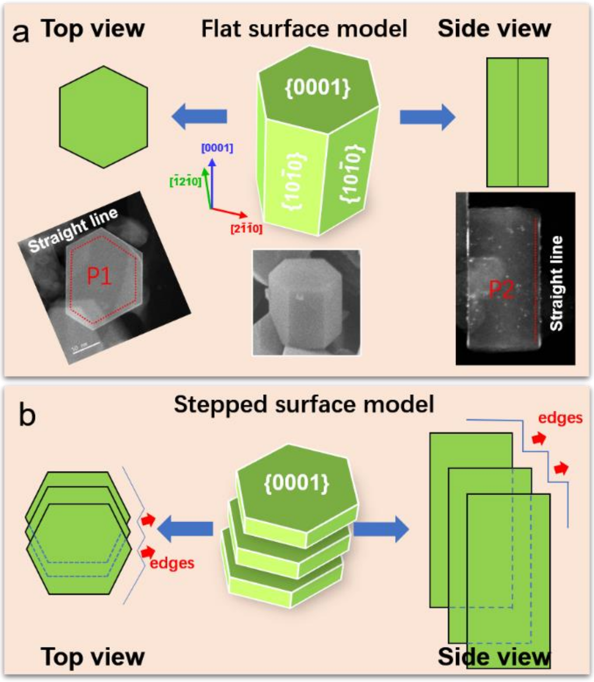
Supplementary Figure 2. HREM (left pane) and iDPC (right panel) image simulation of PTI crystals along the $[2\bar{1}\bar{1}0]$ direction. The simulated HREM and iDPC images along the [0001] and $[2\bar{1}\bar{1}0]$ direction are given in **Supplementary Fig. 1-2**. We also overlap the atomic structure on them. On one hand, clearly, contrast reversal is often observed in the HREM images (see the red arrows and circles in **Supplementary Fig. 1-2**) at some defocus. And when the sample thickness reaches 150 and 200 Å, the HRME images do not intuitively represent the atomic structure (see the yellow arrows in **Supplementary Fig. 1-2**), implying that they are not structure image. On the other hand, there is no evident contrast reversal in the iDPC images when we change the sample thickness or the defocus value (right panel of **Supplementary Fig. 1 and 2**). All the bright spots correspond to atomic column and can be correlated with the atomic potential, albeit blur of images is seen when they are slightly out of focus.



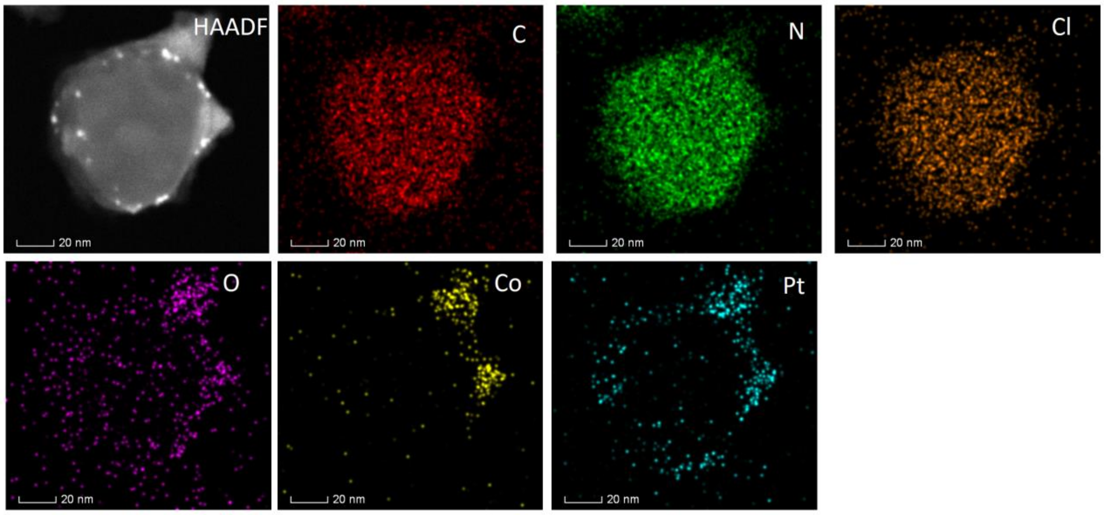
Supplementary Figure 3. Simulated iDPC image of PTI crystals at thickness of 200 Å and defocus value of -100 Å. As a matter of fact, the focal planes of HREM and iDPC images are different. The focal plane of iDPC approximately lie in the center of the sample along the beam direction. Hence, to demonstrate that iPDC image are robust for thicker samples, we simulated iDPC images along the [0001] and [2$\bar{1}\bar{1}$0] directions with a thickness of 200 Å and a defocus value of -100 Å (the half thickness of sample). As presented here, the bright spots correspond to atomic potentials. The image feature match well with the experimental ones in **Fig. 1**.



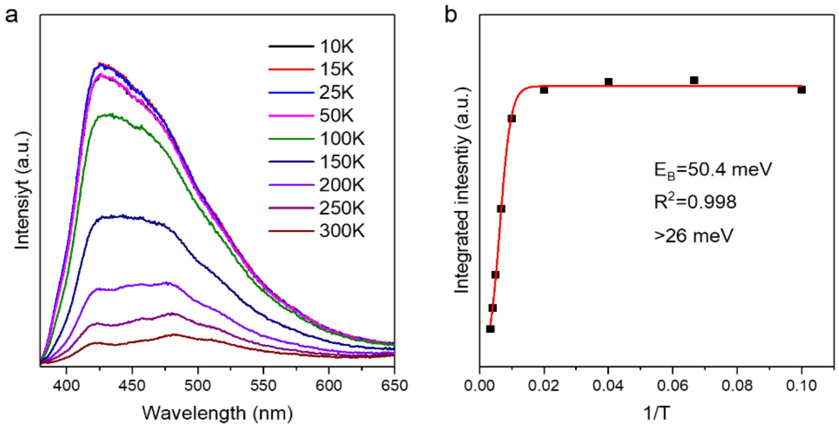
Supplementary Figure 4. Possible configurations of surface H and Li atoms on {10$\bar{1}$0} plane. **a,** monolayer. **b,** bilayer.



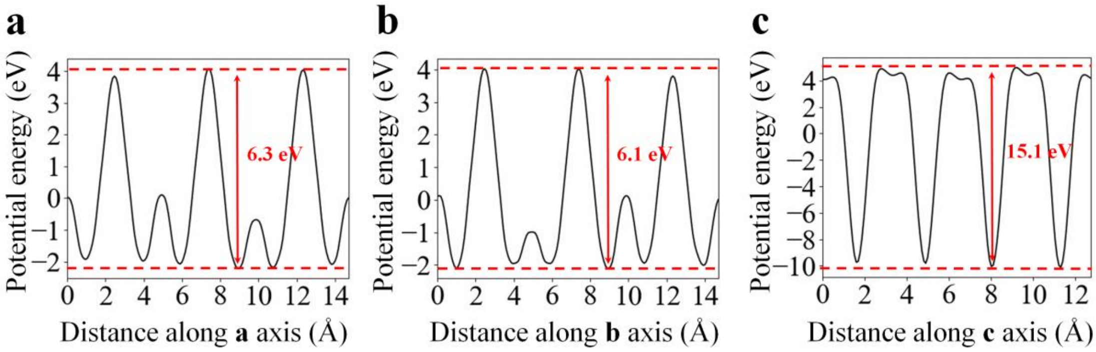
Supplementary Figure 5. Schematic illustration of the flat surface model (**a**) and the stepped surface model (**b**). Because that TEM image is a projection of samples, we need to examine the PTI crystals along two main directions (top view and side view) to identify the surface structures. The two models exhibit different image features in the projected profile. Along the [0001] direction, all the $\{10\bar{1}0\}$ planes are projected as straight lines for our model whilst they are in the form of stepped lines in the stepped surface model (see the red arrows on edges). Two typical examples of PTI crystals (P1 and P2) along the [0001] and $[2\bar{1}\bar{1}0]$ directions are given. Along the [0001] direction, six $\{10\bar{1}0\}$ planes are projected as straight lines without steps (P1), supporting our model. Along the $[2\bar{1}\bar{1}0]$ direction, two $\{10\bar{1}0\}$ planes are projected as straight lines (P2), which is in favor of our flat surface model and not consistent with the stepped surface model.



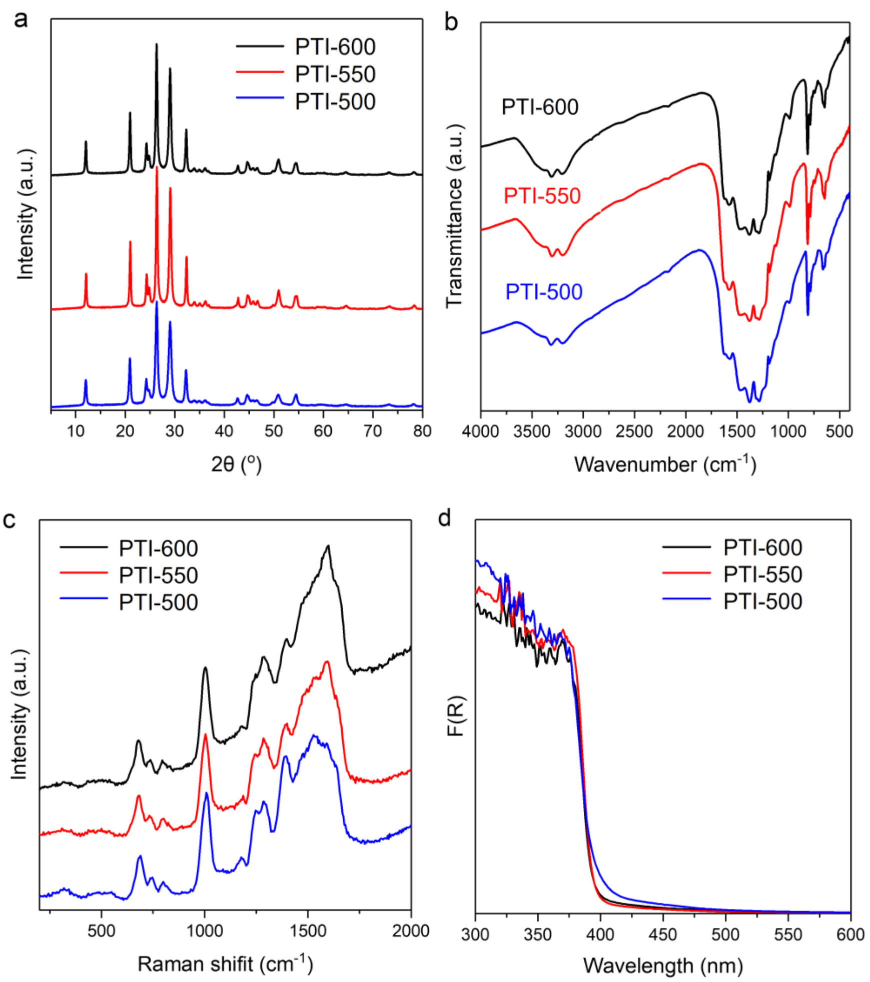
Supplementary Figure 6. EDS maps of a PTI crystal photo-deposited with Pt and Co co-catalyst. Note that the crystal is aligned close to the [0001] direction and hence the side faces correspond to $\{10\bar{1}0\}$ facets. Pt particles are in the form of nano-sized particles (see the brightest particles) whilst the Co particles are large. They all prefer to stay on $\{10\bar{1}0\}$ planes.



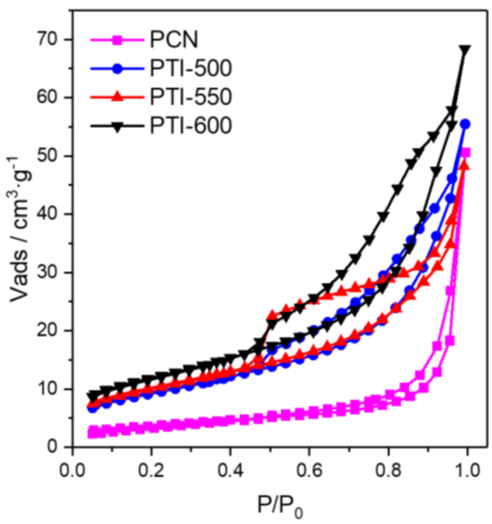
Supplementary Figure 7. a, TD-PL spectra of PTI/Li^+Cl^- under 350 nm excitation. **b,** Integrated PL emission intensity of PTI-550 as a function of temperature from 10 to 300 K.



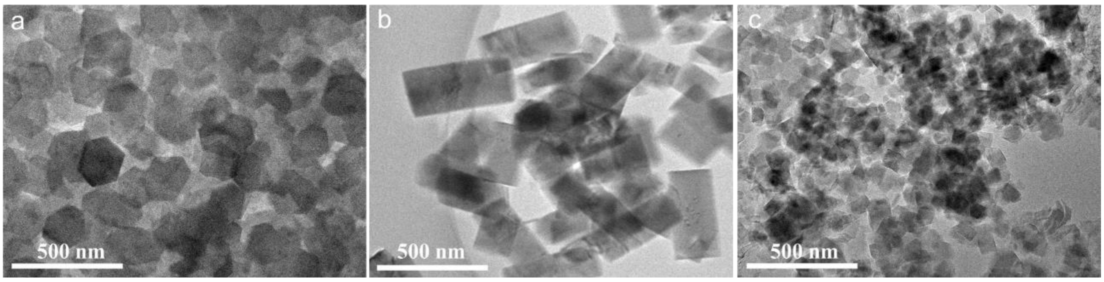
Supplementary Figure 8. The calculated electrostatic potentials of PTI/$Li^{+}Cl^{-}$ crystal along **a**, **b**, and **c** axis directions.



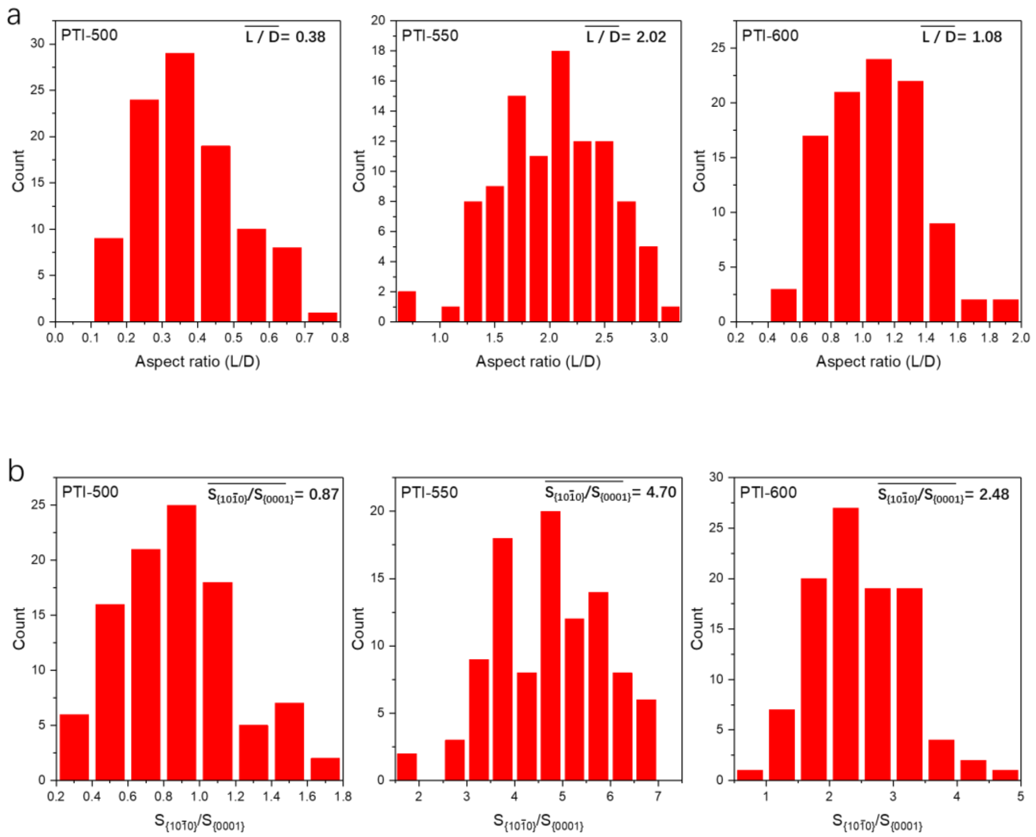
Supplementary Figure 9. Characterization of the PTI samples. **a,** PXRD patterns. **b,** FTIR spectra, **c,** Raman spectra and **d,** UV-vis DRS spectra.



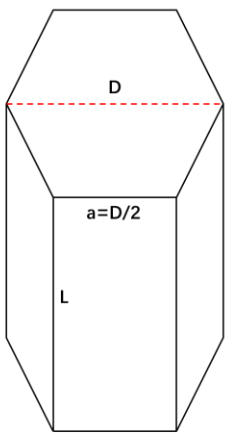
Supplementary Figure 10. N_2-sorption isotherms collected at 77 K.



Supplementary Figure 11 TEM images of **a,** PTI-500, **b,** PTI-550 and **c,** PTI-600.



c



$$S_{\{10\bar{1}0\}} = \frac{D}{2} \times L \times 6$$

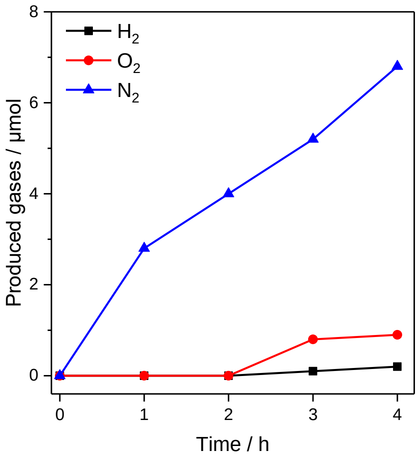
$$S_{\{0001\}} = \frac{1}{2} \times \frac{D}{2} \times \frac{D}{2} \sin 60^o \times 6 \times 2$$

$$= \frac{1}{2} \times (\frac{D}{2})^2 \times \frac{\sqrt{3}}{2} \times 6 \times 2$$

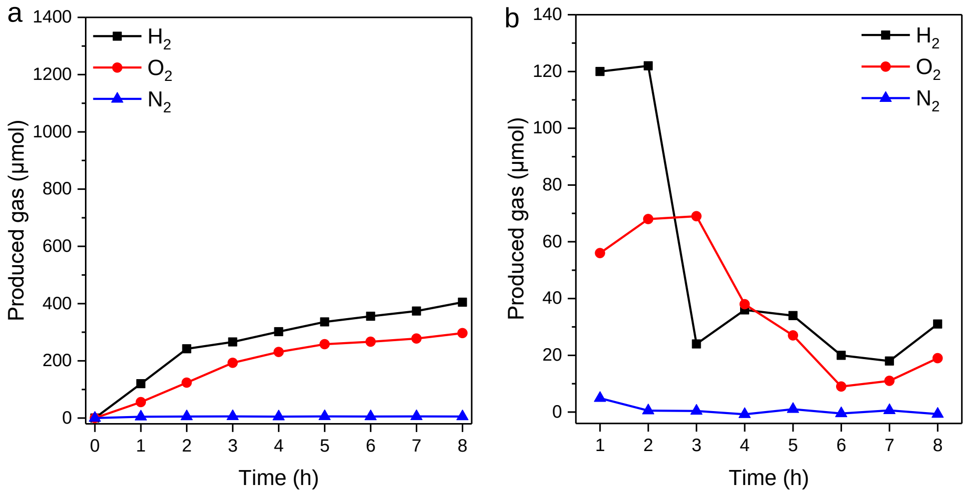
$$= \frac{3\sqrt{3}}{2} \times (\frac{D}{2})^2 \times 2$$

$$\frac{S_{\{10\bar{1}0\}}}{S_{\{0001\}}} = \frac{\frac{D}{2} \times L \times 6}{\frac{3\sqrt{3}}{2} \times (\frac{D}{2})^2 \times 2} = \frac{4}{\sqrt{3}} \times \frac{L}{D}$$

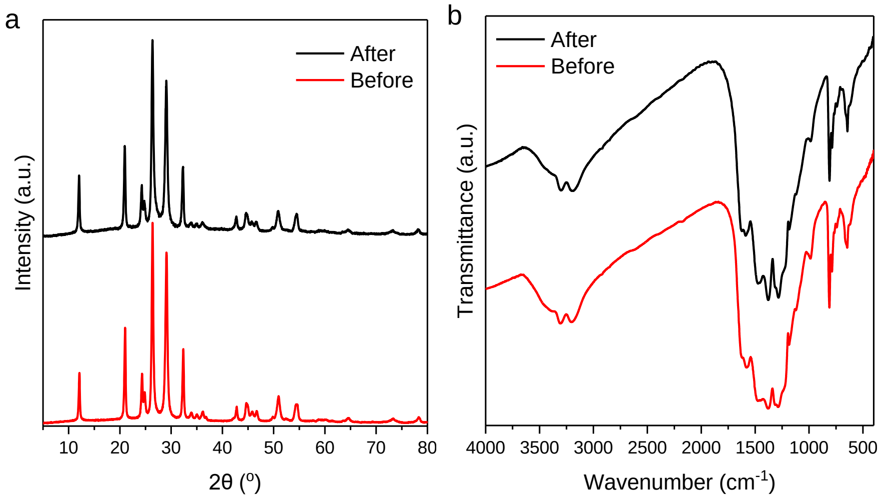
Supplementary Figure 12. Statistical data of aspect ratios (**a**) and $S_{\{10\bar{1}0\}}/S_{\{0001\}}$ (**b**) of PTI at different temperatures. The formula deduction of the correlation between $S_{\{10\bar{1}0\}}/S_{\{0001\}}$ and aspect ratio (L/D) (**c**).



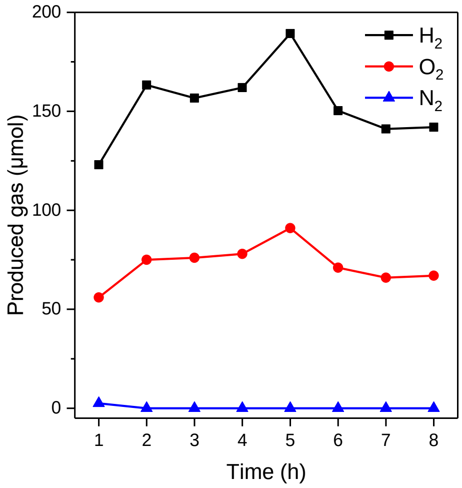
Supplementary Figure 13. The produced gases by PTI-550 crystals without loading co-catalysts.



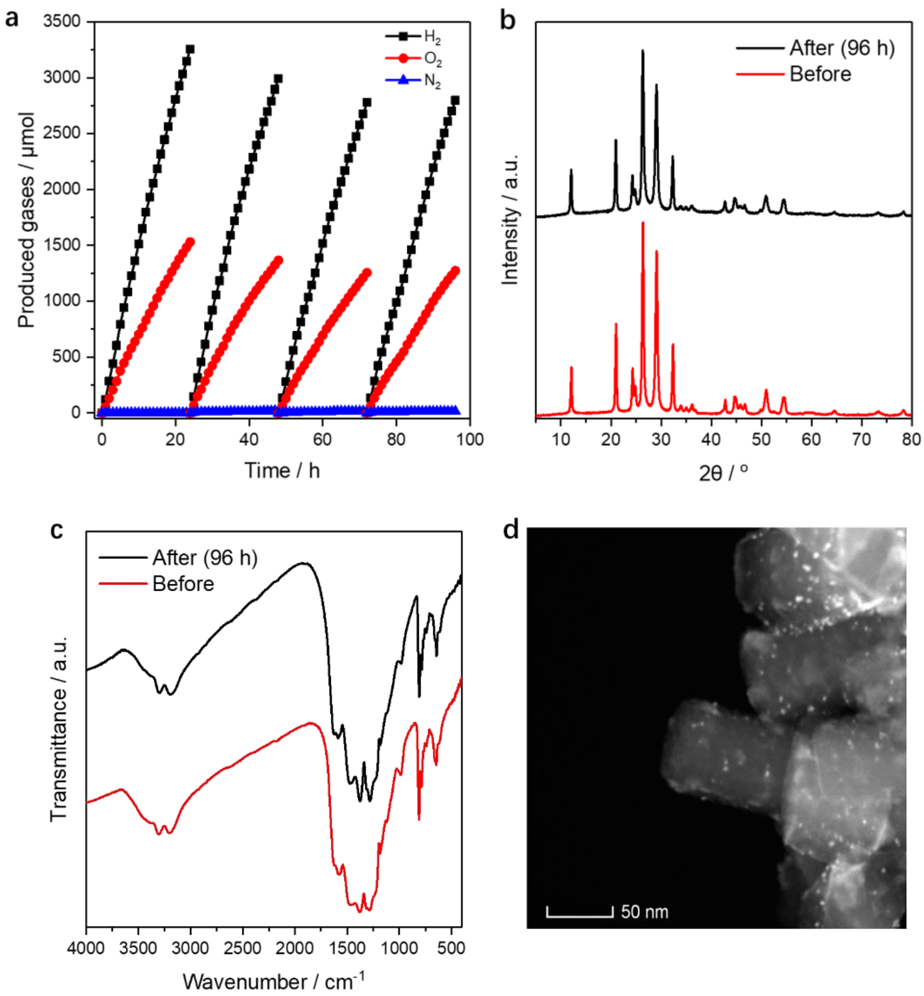
Supplementary Figure 14. a, The produced gases of PTI-550 with prolong time. **b,** The produced gases of PTI-550 in each hour.



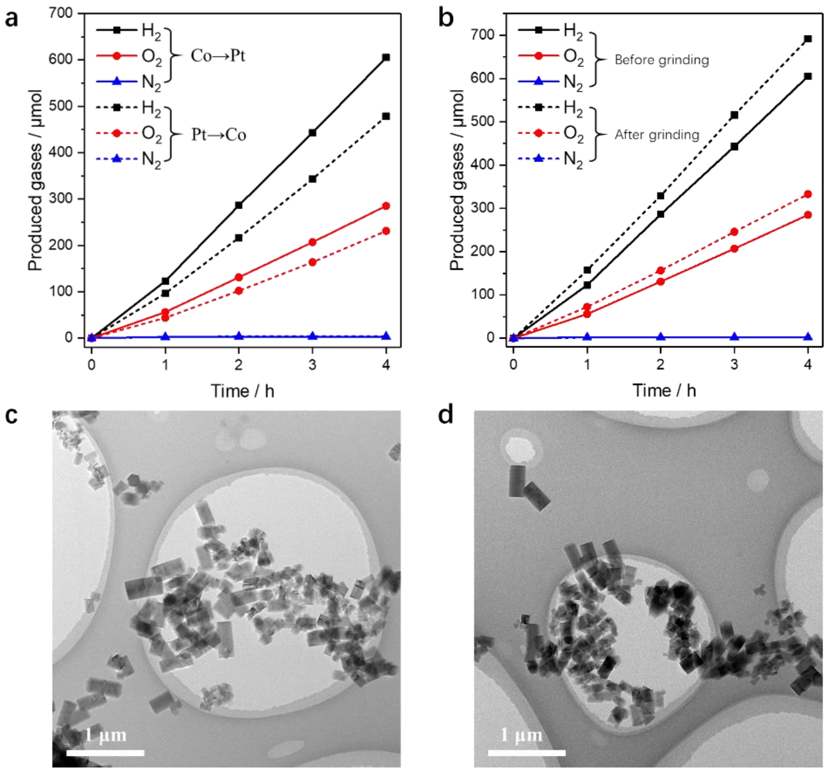
Supplementary Figure 15. a, XRD patterns and **b,** FTIR spectra of PTI-550 before and after 8 hours photoreaction.



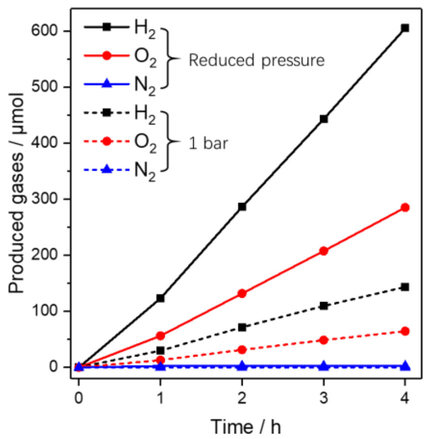
Supplementary Figure 16. The produced gas of PTI-550 in each hour with prolong time (evacuated in each hour after sampling).



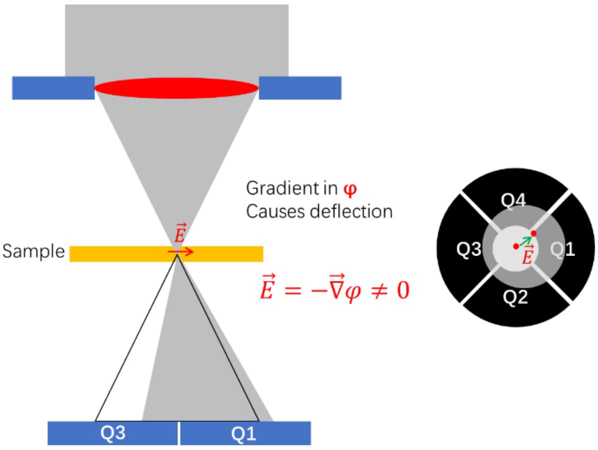
Supplementary Figure 17. a, The accumulated amount of the produced gases by PTI-550 in 96 hours (evacuated every hour after sampling). **b,** XRD, **c,** FT-IR and **d,** morphology of PTI-550 before and after 96 h photoreaction.



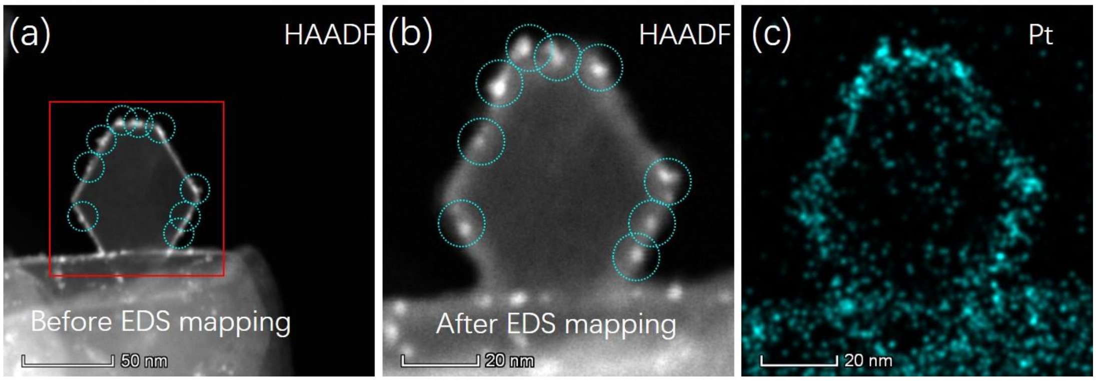
Supplementary Figure 18. a, The accumulated amount of the produced gases by PTI-550 with reversed co-catalysts loading sequence (evacuated every hour after sampling). **b,** The accumulated amount of the produced gases by PTI-550 before and after grinding (evacuated every hour after sampling). The TEM images of PTI-550 **c,** before and **d,** after grinding. Different from photocatalysts with redox reaction on separated facets[1], reduction and oxidization occur on a same $\{101\bar{1}0\}$ plane for PTI crystals. If we deposited Pt before Co, Pt nanoparticles might be buried by large Co particles. In the case of grinding, TEM confirmed that PTI the crystals did not break into segments, implying that the enhanced performance was primarily due to the physical separation of the agglomerated PTI crystal clusters.



Supplementary Figure 19. The accumulated amount of the produced gases by PTI-550 under reduced pressure and 1 bar of Ar gas (evacuated every hour after sampling).



Supplementary Figure 20. Principle of the iDPC imaging. The iDPC-STEM images were acquired using a 4 quadrant (4Q) segmented detector (Q1-Q4) installed on the Themis Z double Cs-corrected STEM. The collection angle of the 4Q detector ranged from 8 to 30 mrad. Electric field could lead to the shift of the central scattered beam, causing a difference in the signal strength of the perpendicular segments (Q2-Q4, Q1-Q3). The two signals correspond to the electric and magnetic fields in the sample on the nanoscale. When the sample is non-magnetic, these two signals represent the fields in x and y direction of the sample plane (DPCx,y), namely, the projected electric field of the sample. In the last, integration of DPCx,y gives access to the projected potential[2]. The iDPC image is a very good approximation of an ideal center of mass (COM) [3]. The obtained integrated COM or iCOM image is then directly proportional to the phase itself[2]. It is confirmed by our iPDC simulation of PTI crystals (**Supplementary Fig. 1 and 2**). Employing the iDPC technique to investigate the atomic structure of PTI crystals has several compelling merits. First, our PTI crystals are relatively thick along the beam direction. The thinnest region would be in the range of 10-20 nm. It is not exfoliated g-CN with a thickness of several layers. If we utilize the convention TEM technique (either cyro-TEM or TEM equipped with a K2 or K3 direct electron detector camera), image contrast reversals would be introduced (**Supplementary Fig. 1 and 2**). Second, iDPC offers high image contrast, especially suitable for materials composed by light and heavy elements concurrently[2]. Third, it is compatible with low dose imaging. For our case, a beam current as low as 2 pA could give a nice interpretable structure image of PTI images during a scanning time of 45s over an area of 75nm*75nm, that is equivalent to a dose rate of 1000 e/(s*Å^2).



Supplementary Figure 21. HAADF image of a crystal imaged before **(a)** and after EDS mapping **(b)**. Pt particles are outlined by cyan circles. **(c)** Corresponding Pt EDS map. It is for sure that the PTI crystal suffered beam damage that induced amorphization during EDS mapping. **Supplementary Figure 21 a** and **b** compares the HAADF image of a PTI crystal before and after EDS mapping, respectively. Pt particles with bright HAADF contrast are denoted by cyan circles. Two important conclusions could be deduced. First, although PTI crystals suffered amorphization during beam damage, the shape of crystal did not change. No strong volume expansion was detected. Second, the positions of Pt particles did not change with respect to the crystal after EDS mapping. No electron-beam induced sputtering or migration of Pt particles occurred. Overall, we demonstrate that no severe volume expansion or migration of Pt particles was detected during EDS mapping, supporting the conclusion that Pt particles prefer to deposit on the $\{10\bar{1}0\}$ planes.

Supplementary Table 1. Chemical compositions of PTI samples obtained from elemental analysis (C, N, H and Cl) and inductive coupled plasma emission spectrometer analysis (Li), respectively.

Sample / Element	PTI for AC-iDPC (wt%)	PTI-500 (wt%)	PTI-550 (wt%)	PTI-600 (wt%)
C	28.82	30.06	29.61	28.27
N	51.57	52.83	52.09	51.46
H	1.36	1.46	1.32	1.29
Cl	12.27	11.93	12.65	12.24
Li	2.34	2.28	2.51	2.65

Supplementary Table 2. The crystal parameters of PTI calculated from the XRD patterns with hexagonal symmetry of $P6_3mc$ (Space group: 185).

Sample	a / Å	b / Å	c / Å	α / °	β / °	γ / °	V / Å^3
PTI-500	8.46033	8.46033	6.75799	90	90	120	418.91
PTI-550	8.44509	8.44509	6.75362	90	90	120	417.13
PTI-600	8.45311	8.45311	6.75997	90	90	120	418.32

2θ/ °	(hkl)	d	I
12.1	(1 0 0)	7.3083	24.9
21.02	(1 1 0)	4.2229	47.7
24.301	(2 0 0)	3.6596	21.8
24.84	(1 1 1)	3.5815	11.3
26.38	(0 0 2)	3.3758	100
29.08	(1 0 2)	3.0681	84.7
32.379	(2 1 0)	2.7627	36.2
33.98	(1 1 2)	2.6361	2.7
35.059	(2 1 1)	2.5574	2
36.18	(2 0 2)	2.4807	3.9
42.799	(2 2 0)	2.1111	7.1
44.66	(3 1 0)	2.0274	7.5
45.861	(3 0 2)	1.977	2.2
46.701	(3 1 1)	1.9434	3.4
49.838	(4 0 0)	1.8282	1.1
50.96	(2 2 2)	1.7905	12
52.4	(2 1 3)	1.7447	0.8
54.6	(3 2 0)	1.6794	8.4
64.718	(2 1 4)	1.4392	1.8
73.36	(5 1 1)	1.2895	1.6
78.38	(6 0 0)	1.219	2.5

Supplementary References

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2. Lazić, I., Bosch, E. G. & Lazar, S. Phase contrast STEM for thin samples: Integrated differential phase contrast. *Ultramicroscopy* **160**, 265-280 (2016).
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